

Project 2: Titanic Technical Report:

Github Repository: https://github.com/rchivate/DTSC_2301_2302

Introduction:

The sinking of the RMS Titanic on April 15, 1912, remains one of the most iconic maritime disasters in history, resulting in the deaths of over 67% of passengers and crew. While often remembered as a tragedy of "bad luck" following a collision with an iceberg, the survival of those on board was deeply tied to the rigid social structures of the Edwardian era. The disaster provides a unique opportunity for social and predictive analysis. Beyond the tragic loss of life, the event serves as a profound case study in human behavior and systemic inequality. Survival was not a matter of chance; it was dictated by the unwritten code of "women and children first" and the physical barriers that separated First Class luxury from Third Class steerage. Those in the lower decks faced a literal maze to reach safety, while the upper class had direct access to the few lifeboats available. This project explores the Titanic passenger manifest to investigate how these social hierarchies influenced life-and-death outcomes.

Data:

The data for this project came from a data analysis competition hosted on Kaggle. The data present in the dataset is accurately sourced from real data of most of the ticketed passengers on board. Because of this, some of the data such as age is missing from the list. The dataset contains 891 rows, one for each recorded passenger, that includes information about the individual such as gender, age, class, etc. Below is a formal description of the variables:

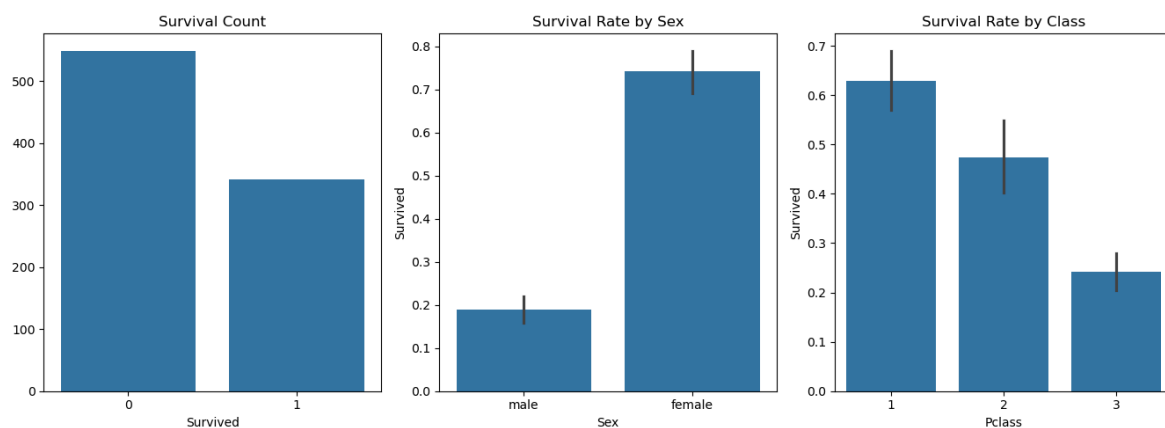
1. **Survived:** Survival (0 = No, 1 = Yes)
2. **Pclass:** Ticket class (1 = 1st, 2 = 2nd, 3 = 3rd)
3. **Sex:** Sex of the passenger
4. **Age:** Age in years
5. **SibSp:** # of siblings / spouses aboard the Titanic
6. **Parch:** # of parents / children aboard the Titanic
7. **Ticket:** Ticket number
8. **Fare:** Passenger fare
9. **Cabin:** Cabin number
10. **Embarked:** Port of Embarkation (C = Cherbourg, Q = Queenstown, S = Southampton)

The dependent variable of this dataset is "Survived", which contains two unique variables "1" for those who survived and "0" for those who did not survive.

Preprocessing and Exploratory Data Analysis:

When looking at the Titanic disaster, it is easy to view it as a series of chaotic, random events. However, my data analysis reveals a much more structured story. The goal of this project was not just to count survivors, but to understand the Social DNA of survival. Raw data often hides the human story. A passenger list gives the names and ages, but the analysis went deeper. I realized that a person's "Title"—whether they were a "Master," "Miss," or "Colonel"—was a powerful proxy for their social standing and the "Women and Children First" protocol. By distilling hundreds of names into these core social categories, I transformed raw text into a

strategic predictor of survival. I discovered that survival wasn't just an individual journey. By combining data on siblings, spouses, and parents into a single "Family Size" metric, I could see a clear pattern: those traveling in small, coordinated groups often fared better than those who were entirely alone or those in excessively large groups who may have struggled to stay together during the evacuation. My initial analysis showed that the data is naturally skewed toward those who did not survive. If I simply guessed that everyone perished, I would be "accurate" most of the time but fundamentally wrong in my mission. To combat this, I used advanced metrics that measure the model's ability to distinguish between classes, rather than just its ability to guess. I have successfully moved beyond simple statistics to create a model that understands the socio-economic and demographic currents that decided a passenger's fate.



(Figure 1: EDA Visualizations)

After preprocessing the data, I made some visualizations to better understand the data that I was working with and understanding the inherent biases that may come out of it (Figure 1). Figure 1A shows that there is more passenger data about the victims rather than the survivors. This means that if the model guessed that everyone died, then the accuracy will still be high enough if the method is not scrutinized. Figure 1B tells us that more females survived by a landslide than males. And to nobody's surprise Figure 1C tells us that the first class, the rich, had significantly more survivors than class 2 or 3.

Modeling:

The data was split into two parts to ensure that the evaluation was unbiased and data was fully randomized: 80% of the data was used to train the model, while the other 20% was used to test.

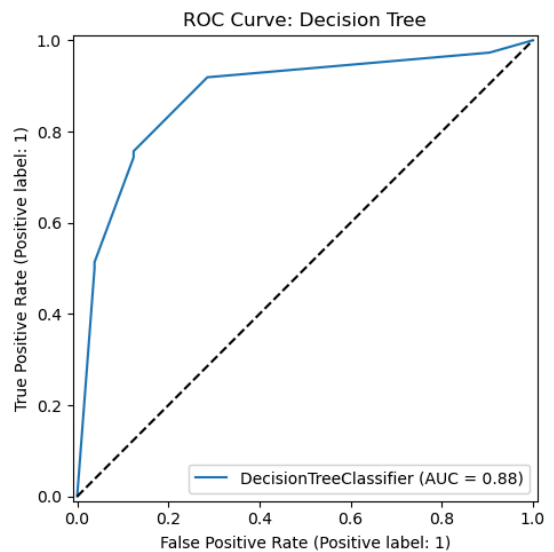
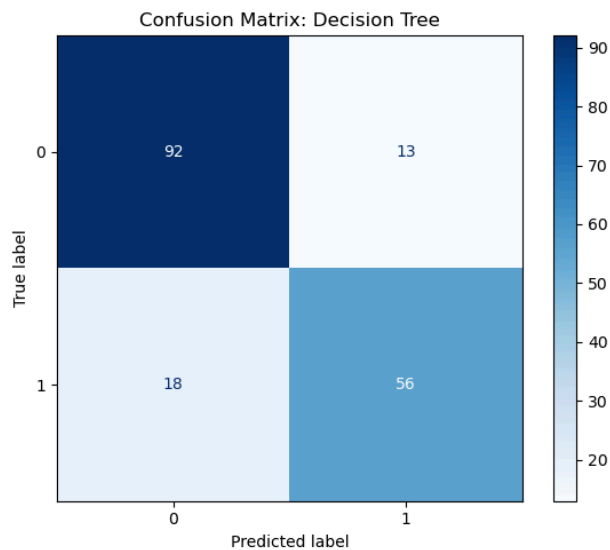
Three predictive models were compared: Decision Tree (with a depth of 4), Random Forest, and Logistic Regression. The performance was primarily evaluated using recall score, precision score, and ROC curve.

Results:

Decision Tree

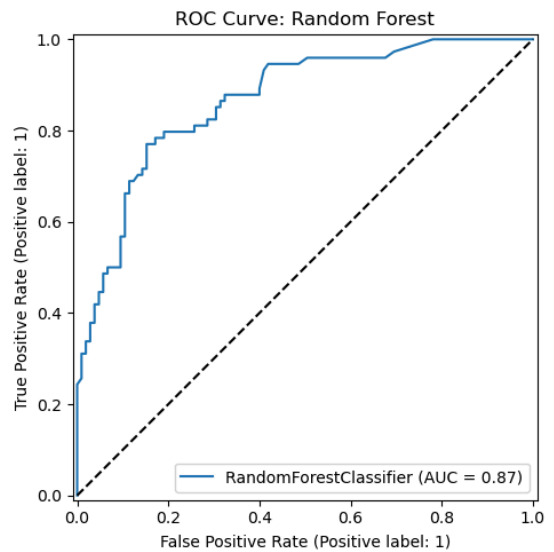
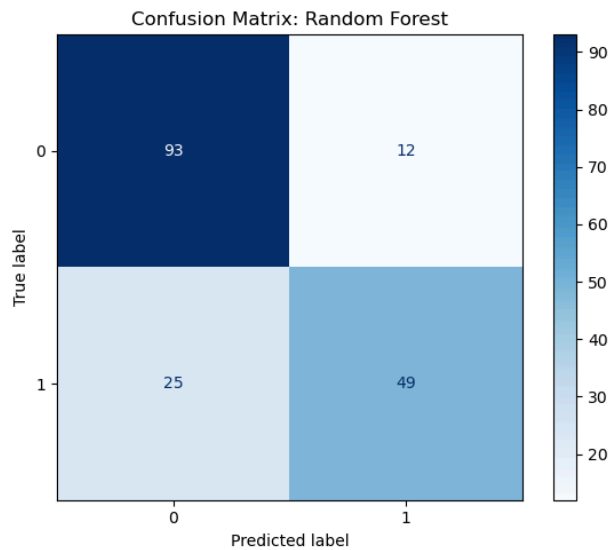
- Score: 0.8268

- Precision: 0.8116
- Recall: 0.7568
- ROC-AUC: 0.8786



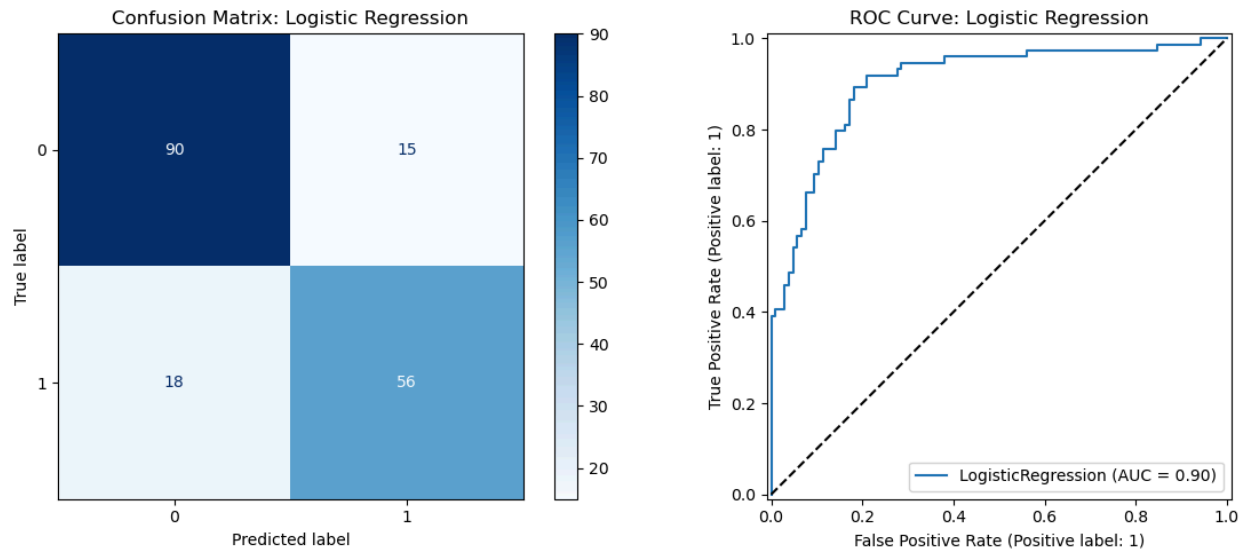
Random Forest

- Accuracy: 0.7933
- Precision: 0.8033
- Recall: 0.6622
- ROC-AUC: 0.8680



Logistic

- Accuracy: 0.8156
- Precision: 0.7887
- Recall: 0.7568
- ROC-AUC: 0.9040



Out of the three models, the Logistic Regression performed the best with the ROC-AUC Score of 0.9040, followed by the Decision Tree (ROC-AUC Score = 0.8786) and Random Forest (ROC-AUC Score = 0.8680).

The ROC-AUC score tells us how good the model is at distinguishing between a survivor and a victim. A high score tells us that the model is not just utilizing luck, but understanding the underlying pattern that may exist within the dataset. The ROC-AUC score ignores the imbalance of the data, since there are more people who perished than survived. If the model simply guessed the passenger was a victim it would be correct most of the time and prove to be a useless model since the guess has no basis.

Even though Logistic Regression had the highest ROC-AUC score, it is not the best model for predicting survivors after looking at the precision and recall scores. The precision score tells us how many actually survived from the survivors that the model predicted. Logistic regression had the lowest precision score out of the three models of a 0.7887. Similar case with the recall score, out of all the actual survivors, how many the model found. Logistic Regression has the lowest recall score of 0.7568.

The best model according to the precision and the recall scores, since they are the most vital in the context of being able to predict who actually survived the Titanic, is the Decision Tree. Random Forest has a high precision score of 0.8033, but a low recall score of 0.6622. The decision tree has a precision score of 0.8116 and a recall score of 0.7568.

Resources:

- <https://scikit-learn.org/stable/>
 - Scikit-learn library documentation
- <https://www.noaa.gov/office-of-general-counsel/gc-international-section/rms-titanic-history-and-significance>

- Background research
- <https://www.kaggle.com/c/titanic/data>
 - Titanic Data Set from Kaggle Competition

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- Gemini and Claude were used to troubleshoot error and warning messages, and correct grammatical errors